finally found one, as engineers of the Assam-based North-east Frontier Railway (NFR) blended existing technology with a twist of *jugaad* to set off alarms in train engines and at stations every time a herd of elephants was found approaching railway tracks. It has come to be called IDS or the Intrusion Detection System.

It was in late 2021 that the Railways chanced upon a solution lying in plain sight: optic fibre cables (OFC) laid along tracks to transmit signals. They only had to incorporate some Distributed Acoustic Sensing (DAS) technology to transform those very OFCs into intelligent sensors capable of detecting acoustic vibrations along their entire length. All one had to do was send a laser pulse and analyse the scattered light patterns resulting from the vibrations and disturbances in the cable.

If the signal bounced back, it meant the cable's continuity was broken—a phenomenon called 'backscatter'. By monitoring these disturbances, the DAS, working through devices attached to the OFCs, could pinpoint the location, direction and even the type of intrusion. The system was then connected to a central server where an Artificial Intelligence programme constantly learned to distinguish the soundwave signature of elephants crossing or approaching tracks from everything else. "The soundwave signature of elephants is different from that of humans," says Upendra Bhumij, principal chief signalling and telecom engineer of NFR. "The machine had to learn it and trigger an alarm only when elephants were approaching to reduce the chances of false positives."

The sensitivity of the OFCs was tweaked so that they could detect the soundwave of elephants even when the animals were up to 80 metres away from the track. The central server can set off elephant alerts at train control offices, level-crossing gates and offices of the station masters in real time. Alarm machines have also been fitted inside engine cabs. Trains without it are alerted by the nearby station master over walkie-talkie.

From November 2022 to March 2023, the system was installed to cover six elephant corridors along the tracks between Lumding and Lanka in Assam and eight corridors between Hasimara and Chalsa in the Alipurduar division in West Bengal covering 140 kilometres. "Not a single collision with, or death of, elephants happened in those stretches since the trial started last year," Bhumij says. Indeed, in the first six months, there were 590 instances of elephants being saved from collisions with trains, as drivers, warned by IDS, applied brakes or drove at slower speeds. Now, an additional 400 km of tracks are being brought under the ambit of the IDS in Assam and North Bengal.

R.K. Mondal, a loco pilot of freight trains in Lumding, Assam, recalls how the new technology saved the situation in an elephant corridor once. "I slowed down the train after getting a signal. After some time, I saw elephants crossing the tracks peacefully," he says. Now, NFR gets 50 elephant alarms every day in the stretches where the system is installed. "For us, each of these alarms represents a potential collision avoided," says NFR spokesman Sabyasachi De. The Railways has decided to instal the system along over 700 km of tracks in East Coast Railways, Southern Railways, Northern Railways and South Western Railways, where trains pass through known elephant corridors.

"The government has asked the highways authority to use the IDS to save elephants. We are also exploring other use cases in the Railways, like getting alarms before landslides or river erosion," says Anshul Gupta, adviser, RailTel and former general manager of NFR, during whose tenure the IDS was started.

For a country that has declared the elephant its 'national heritage animal', India has had an inordinately high number of elephant deaths due to unnatural causes. Hopefully, with the IDS in place, one man-made reason may be off the list forever. ■

HAPPINESS QUOTIENT

- **The Intrusion Detection System uses Distributed Acoustic Sensing technology to send laser signals along optic fibre cables, and learn soundwave signature of elephants to caution trains/ stations**
- **Used in railway tracks within 14 elephant corridors in Assam and Bengal, they have been a great success. Railways will instal IDS along over 700 km of tracks in three other zonal railways**